

Aggression Analysis in YouTube News Comments

Team:- Neuronauts

Abstract—

1.

This report analyzes aggression patterns in YouTube comments on news videos related to the Venezuelan political crisis and conflict-driven events. The study examines how audience behavior varies across multiple dimensions, including **video-based**, **time-based**, **gender-based**, and **location-based** perspectives.

Videos are categorized into **General/Neutral News**, **Political News**, and **War-related News**, and comments are classified as **Aggressive**, **Calm/Peaceful**, or **Neutral**. Word-cloud visualization and word-frequency scoring are used to identify dominant vocabulary and aggression-related language. **Time-based analysis** compares early and late comment phases, while **gender- and location-based analyses** explore differences in participation and emotional intensity.

Findings are interpreted using the **ABC (Antecedent-Behavior-Consequence)** framework, showing that war-related content generates higher aggression, whereas general news remains comparatively stable, with aggressive comments forming a minority but exerting significant influence on discussion dynamics.

2. VIDEO BASED ANALYSIS

Akshat Garg(230096)

The video-based analysis shows that **war-related news generates the highest level of aggressive comments**, followed by political news, while general/neutral news remains the most stable discussion environment. Although aggressive comments form a minority across all categories, their proportion increases significantly with conflict-heavy framing. Overall, the findings indicate that **content intensity directly influences online aggression**.

2.1. Methodology

2.1.1. Video Categorization

Videos were categorized into three types:

- **Politics:** leadership/government-oriented topics (e.g., president, regime, diplomacy).
- **War:** military action, conflict, violence, threats, armed escalation.
- **General/Neutral News:** reporting-focused, informational updates without heavy war/political conflict framing.

2.1.2. Comment Tone Classification

Each comment was labeled into one of three tones:

- **Aggressive:** hostile, blame-oriented, angry expressions.
- **Calm/Peace:** supportive, empathy-based, peaceful reasoning.
- **Neutral:** informational, normal opinion, fact-sharing, low emotional intensity.

2.1.3. Word Score + Word Cloud

A word frequency score was computed based on:

$$\text{score}(\text{word}) = \frac{\text{count}(\text{word})}{\text{total comments}}$$

Word clouds were created using the most frequent words within each category and tone.

2.2. Overall Dataset Analysis

2.2.1. Overall Tone Distribution

The full dataset contains the following tone distribution:

Table 1. Overall tone distribution (all categories combined)

Tone	Count	Percentage
Neutral	6447	52.2%
Calm	5398	43.7%
Aggressive	517	4.2%

2.2.2. Overall Word Cloud (Aggression-Focused Interpretation)

The overall word cloud is dominated by high-frequency terms such as *people*, *venezuelans*, *maduro*, *president*, *criminals*, *media*, which confirms that the discussion is strongly centered on the Venezuelan crisis and leadership-related events. However, from an **aggression perspective**, the most important observation is the presence of direct blame and harm-related words like *criminals*, *torture*, *prison*, *regime*, *die*, *suffer*. These words indicate that a portion of users are not only reacting to updates, but also expressing strong anger toward the situation and the actors involved.

At the same time, calmer vocabulary such as *rights*, *justice*, *innocent*, *medical*, *care*, *attention*, *pressure*, *release* appears frequently, suggesting that even when aggression exists, many users frame their response around awareness, humanity, and accountability rather than personal attacks. Overall, the word cloud shows a **mixed comment environment**: aggression is visible through harsh terms, but the dominant engagement still revolves around issue-based discussion.



Figure 1. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

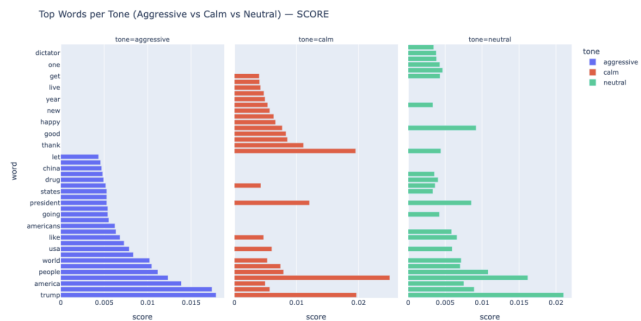


Figure 2. Word frequency score plot showing the most repeated words in comments across Aggressive, Calm, and Neutral tones.

2.2.3. ABC Model (Overall: Aggression-Centered)

A (Antecedent): The antecedent is exposure to crisis-driven news content involving *maduro*, *president*, *regime*, *country* and public suffering. This creates strong emotional triggers because viewers interpret the situation through justice and harm framing (*criminals*, *torture*, *prison*, *die*, *suffer*).

B (Behavior): The behavioral outcome is visible through comment tone. Most viewers remain **neutral or calm** by discussing the issue using words such as *people, media, rights, justice, innocent, medical, care*. Aggressive behavior emerges when users shift into blame and anger language, reflected by repeated harsh vocabulary like *criminals, torture, prison, regime*.

C (Consequence): The consequence of aggression is increased polarization in comment threads. When aggressive words dominate, discussions can become hostile and identity-targeted, reducing constructive debate. However, because neutral and calm comments still dominate overall, aggression remains a **smaller but impactful layer** of the discussion, mainly intensifying conflict rather than controlling the entire conversation.

2.3. General / Neutral News Analysis

2.3.1. Tone Distribution (General News)

Table 2. General News tone counts

Tone	Count	Percentage
Neutral	767	55.6%
Calm	557	40.4%
Aggressive	56	4.1%

2.3.2. Word Cloud Observation (General News: Aggression-Focused)

The General/Neutral news word cloud is dominated by terms such as *maduro, venezuelan, people, president, media*, showing that viewers are mainly reacting to leadership-related updates and public-impact discussion. From an **aggression perspective**, the key words that highlight anger and blame are *criminals, dictator, prison, tortured, arrested*. These words indicate that a small portion of users interpret the situation as injustice and respond with strong negative emotion.

At the same time, calmer issue-framing words like *rights*, *attention*, *pressure*, *release* appear frequently. This suggests that in general news, viewers mostly express concern in a controlled way, focusing on awareness and accountability rather than turning the comment section into direct conflict.



Figure 3. General News word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

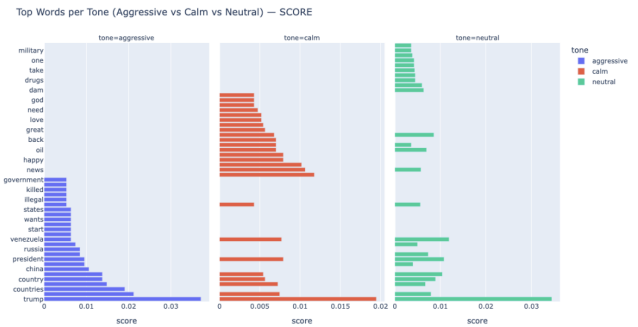


Figure 4. Word frequency score plot showing the most repeated words in comments across Aggressive, Calm, and Neutral tones.

2.3.3. ABC Model (General News: Aggression-Centered)

A (Antecedent): The antecedent is exposure to informational news updates about leadership and public issues, represented by words like *maduro*, *president*, *venezuelan*, *people*. This creates awareness and judgement-based reactions.

B (Behavior): The dominant behavior stays **neutral or calm**, shown by controlled vocabulary such as *people, media, rights, attention*. However, aggression emerges when users use harsh blame words like *criminals, dictator* and harm-related words like *prison, tortured, arrested*.

C (Consequence): Because general news has low aggression overall, discussions remain mostly stable. However, aggressive vocabulary can still trigger hate/blame threads and reduce constructive engagement, even if it represents a smaller portion of comments.

2.4. Politics News Analysis

2.4.1. Tone Distribution (Politics)

Table 3. Politics tone counts

Tone	Count	Percentage
Calm	3740	50.2%
Neutral	3477	46.6%
Aggressive	239	3.2%

2.4.2. Word Cloud Observation (Politics: Aggression-Focused)

The Politics word cloud is dominated by words such as *maduro*, *president*, *people*, *venezuelans*, *country*, *media*, showing that viewers are actively reacting to leadership-based news and national-level decisions. From an **aggression perspective**, political content still contains noticeable blame framing through repeated terms like *criminals*, *dictator*, *regime*, *prison*, *tortured*, *arrested*. These words represent anger and frustration directed towards the system or leadership.

However, politics also shows strong calm/neutral issue framing through words like *rights*, *attention*, *pressure*, *release*. This indicates that many viewers try to discuss politics in a controlled and mature way, but aggressive comments emerge when users interpret the situation as injustice or abuse of power.

Top Words per Tone (Aggressive vs Calm vs Neutral) – SCORE

tone=aggressive

tone=calm

tone=neutral

word

score

score

score

tone

- aggressive
- calm
- neutral

Tone	Count	Percentage
Neutral	2203	62.5%
Calm	1099	31.2%
Aggressive	222	6.3%

[illegible]

3

2.6.1. Aggression Comparison Across Categories

Aggressive comment counts in each category are:

Table 5. Aggressive comment comparison across categories

Category	Aggressive Count	Aggressive %
General / Neutral News	56	4.1%
Politics	239	3.2%
War	222	6.3%

Key Observation: Even though Politics has a slightly higher count of aggressive comments than War, the **war category has the highest aggressive percentage (6.3%)**, meaning aggression is more intense and more frequent relative to total comments in war videos.

2.6.2. Why War Generates More Aggressive Comments

War content naturally carries high emotional intensity and includes strong harm-related framing such as *criminals*, *torture*, *prison*, *regime*, which increases anger-driven responses. General news videos are more informational and stable, and politics often encourages mature debate, which helps keep aggression lower.

2.6.3. ABC Model (Cross-Category: Aggression Explanation)

A (Antecedent): The antecedent is the type of content shown in the video. War videos act as a high-emotion trigger because they are directly associated with crisis and suffering, which increases sensitivity and anger. Political videos also trigger strong opinions, but many viewers attempt controlled discussion. General news videos are mainly informational and create the lowest conflict environment.

B (Behavior): When the content is more emotionally aggressive (war framing), users are more likely to post aggressive comments. This explains why war videos show the highest aggressive proportion, while politics and general news show relatively lower aggression. In general news and politics, more users remain calm or neutral, whereas in war, more users shift toward blame and hostility.

C (Consequence): The consequence is that war-related comment sections become more polarized and toxic compared to other categories. Aggressive language spreads faster in war discussions, creating negative feedback loops where one aggressive comment triggers replies and escalations. Meanwhile, general news remains the most stable discussion environment due to low aggression, and politics stays relatively balanced because calm/neutral engagement dominates.

2.7. Approach (How the Analysis Was Done)

To study aggression in YouTube comments related to Venezuelan conflict and political news, videos were first grouped into three categories: **General/Neutral News**, **Politics**, and **War**. Comments were collected and pre-processed by removing unnecessary symbols, links, and stopwords to focus only on meaningful words.

Each comment was then classified into one of three tones: **Aggressive**, **Calm/Peace**, or **Neutral**. After classification, word frequency scores were calculated using word count divided by total comments, and word clouds were generated to visualise the most repeated terms in each category. Finally, an **ABC (Antecedent–Behavior–Consequence)** framework was used to explain why aggressive reactions increase in war-heavy content compared to general news and politics.

2.8. Conclusion

This study demonstrates that YouTube comment tone strongly depends on the type of content. While the majority of comments remain

neutral or calm, war-related content increases aggression due to emotionally triggering vocabulary such as torture, prison, and regime. Politics encourages relatively calm discussion, while general news remains the most stable environment. This analysis highlights how word cloud patterns and tone scores can effectively explain online audience behavior across different news themes.

3. TIME-BASED COMMENT ANALYSIS

Chirag Goyal (230335)

To observe how aggression changes as the comment section evolves, the dataset was split into three time-based segments: **First 500 comments** (early engagement), **Last 500 comments** (late engagement), and the **Overall dataset**. Each segment is analysed using **tone distribution** and **word cloud patterns** to capture aggression intensity over time.

3.1. Overall Tone Distribution (Full Phase)

The overall dataset contains the following tone distribution:

Table 6. Overall tone distribution (Full dataset)

Tone	Count	Percentage
Neutral	1523	64.3%
Calm	657	27.7%
Aggressive	183	7.7%

3.2. Overall Word Cloud (Aggression-Focused Interpretation)

The overall word cloud is dominated by high-frequency terms centered around war and global conflict framing. Words such as *war*, *russia*, *america*, *attack*, *china* indicate that viewers strongly associate the discussion with aggression-driven geopolitical conflict. From an **aggression perspective**, harsh vocabulary such as *attack*, *kill*, *invade* reflects anger and blame reactions toward the situation and actors involved. At the same time, calmer reasoning words like *peace* also appear, indicating that although aggression is visible, some viewers attempt controlled discussion. Overall, the dataset shows a **mixed environment**: neutral discussion dominates, but aggression forms a strong and emotionally intense layer.



Figure 9. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

3.3. Early Phase: First 500 Comments

Table 7. Tone distribution (First 500 comments)

Tone	Count	Percentage
Neutral	304	60.8%
Calm	156	31.2%
Aggressive	40	8.0%

The first 500 comments represent the **initial reaction stage**, where users respond quickly to the content before long discussion chains form. The early-phase word cloud contains direct and emotionally intense vocabulary, including strong conflict framing and hostile terms. From an **aggression perspective**, this indicates that early viewers react impulsively with anger and blame-driven language, leading to a slightly higher aggressive proportion (8.0%). However, neutral and calm comments still dominate, showing that many users engage through opinion-sharing and discussion rather than direct hostility.

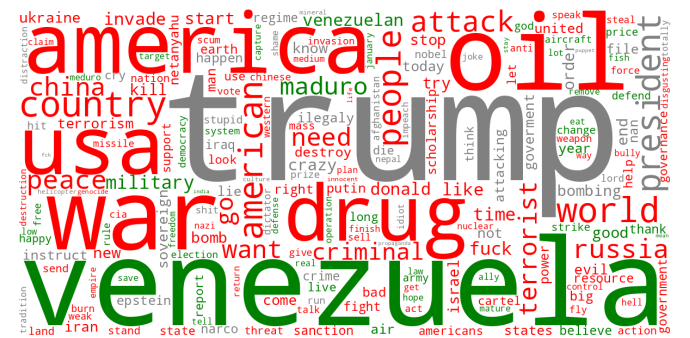


Figure 10. First 500 word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

3.4. Late Phase: Last 500 Comments

Table 8. Tone distribution (Last 500 comments)

Tone	Count	Percentage
Neutral	357	71.4%
Calm	105	21.0%
Aggressive	38	7.6%

The last 500 comments represent the **late engagement stage**, where discussion continues after the video has already gained broader exposure. Late-stage engagement is usually shaped by **reply chains, repetition, and polarization**, where users often respond to earlier comments instead of only reacting to the video. Compared to the early phase, the late phase shows a higher proportion of **neutral comments** (71.4%), while aggression slightly decreases (7.6%). This suggests the discussion becomes more stable and informational over time, even though aggression remains present through repeated blame and conflict-related framing.

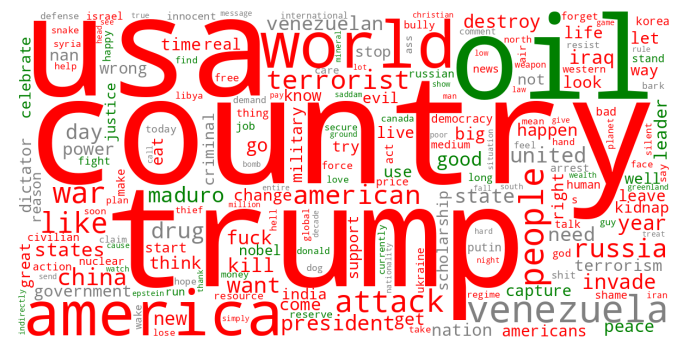


Figure 11. Last 500 word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

3.5. Final Time-Based Comparative Insight

The time-based comparison shows that aggression changes slightly across time:

- **First 500 (Early Phase):** Aggression is highest (8.0%), showing stronger impulsive emotional reactions.

- **Last 500 (Late Phase):** Aggression decreases slightly (7.6%), while neutral discussion increases (71.4%).
- **Overall:** Aggression remains a **smaller but impactful layer** of the full discussion environment (7.7%).

3.6. ABC Model (Time-Based: Aggression-Centered)

A (Antecedent): The antecedent is exposure to war/political conflict framing, which creates strong emotional triggers such as anger, blame, and moral judgement. In the early phase, this trigger is stronger because viewers react instantly with shock and hostility. In the late phase, the antecedent shifts into discussion-based triggers, where users respond to repeated narratives and earlier threads.

B (Behavior): The behavioral outcome is visible through tone distribution. Aggression appears slightly higher in the early phase (8.0%) due to direct emotional reaction and harsh vocabulary. In the late phase, most comments shift toward **neutral discussion** (71.4%), while aggression remains present but slightly reduced (7.6%), indicating stabilization over time.

C (Consequence): The consequence of aggression is escalation and polarization in comment threads. Early aggression can trigger toxic chains and hostility, while late-phase aggression contributes to repeated blame narratives and debate cycles. However, because neutral and calm comments dominate across both phases, aggression remains influential but does not fully control the overall conversation environment.

3.7. Conclusion

This study presented a **time-based aggression analysis** of YouTube comments by comparing the **First 500 (early phase)**, **Last 500 (late phase)**, and the **overall dataset**. The results show that aggression is present across all phases, but it remains a **minority layer** compared to neutral discussion. The early phase shows the **highest aggression (8.0%)**, indicating stronger impulsive emotional reactions immediately after video exposure. In contrast, the late phase shows a **higher neutral share (71.4%)** and a slight reduction in aggression (7.6%), suggesting that discussions become more stable and information-driven over time.

Overall, the findings highlight that **aggression intensity slightly decreases as comment sections mature**, while neutral engagement increases. This supports the idea that initial exposure triggers immediate anger and blame responses, whereas later engagement is shaped more by reply chains and repeated debate cycles. Therefore, time-based segmentation provides a clearer understanding of how online comment aggression evolves and how a relatively small aggressive group can still influence the emotional tone of the overall conversation environment.

4. GENDER-WISE SENTIMENT ANALYSIS

Muskan Yadav (230673)

To examine gender-based differences in engagement, the dataset was segmented according to the **gender of the commenter**, namely **male, female, and unknown** categories. Each segment is analysed using **sentiment distribution (aggressive, neutral, calm)** and **word cloud patterns** to identify differences in emotional expression, engagement intensity, and dominant themes across genders.

4.1. Overall Tone Distribution (Full Phase)

Table 9. Gender-wise tone distribution (Full dataset)

Gender	Aggressive	Neutral	Calm	Total
Female	165	69	27	261
Male	2768	1213	410	4391
Unknown	48	25	10	83
Total	2981	1307	447	4735

4.2. Overall Word Cloud Analysis

The overall word cloud presents a consolidated view of dominant themes across aggressive, neutral, and calm comments related to world-at-war content on short video platforms. Aggressive terms (shown in red) such as *world*, *trump*, *venezuela*, *people*, *war*, *attack*, *criminal*, and *terrorist* appear most prominently, indicating that discussions are largely driven by emotionally charged language and conflict-centric narratives. Neutral terms (in gray), including *president*, *news*, *state*, *power*, *regime*, and *military*, reflect informational or descriptive engagement without strong emotional expression, while calm terms (in green) such as *peace*, *help*, *good*, and *thank* appear less frequently, suggesting limited positive or reflective discourse. Overall, the dominance of aggressive vocabulary highlights a strong skew toward reactive and emotionally intense engagement, supporting the observation that short-form war-related content tends to amplify negative emotional responses over calm or neutral cognitive processing.



Figure 12. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

4.3. ABC Model (Gender Based Analysis)

A (Antecedent): The primary antecedent in this analysis is exposure to world-at-war related content through short-form video platforms such as YouTube Shorts and Instagram Reels. The repeated appearance of such content in users' feeds, driven by platform algorithms and ongoing global events, acts as a common emotional trigger across genders. Additional antecedents include emotional arousal, curiosity, and the urge to express opinions in response to rapidly consumed war-related information.

B (Behavior): The observed behavior manifests as active commenting on war-related short videos, with clear gender-based differences in participation levels. Male users demonstrate higher overall engagement, reflected in a significantly larger number of comments, while female users contribute fewer comments. Across both genders, the predominant behavioral response involves negative or aggressive commenting, indicating emotionally charged engagement, whereas positive commenting behavior remains comparatively limited.

C (Consequence): The consequence of this engagement pattern is a dominance of negative emotional expression within online discussions surrounding war-related short-form content. The higher volume of negative comments among male users suggests greater exposure or

sustained engagement, potentially leading to increased emotional involvement and cognitive load. For both genders, the relative scarcity of positive responses indicates that short-form war content is more likely to reinforce stress, anxiety, or emotional strain rather than promote positive cognitive outcomes.

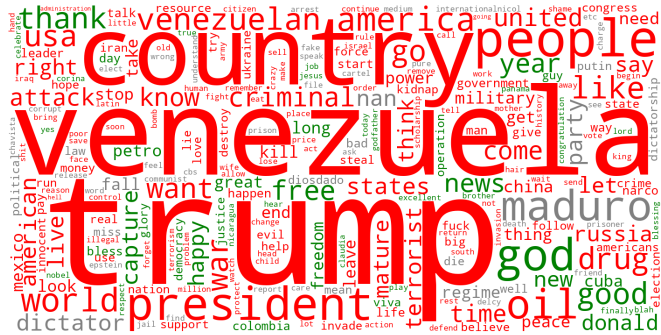


Figure 13. Males word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

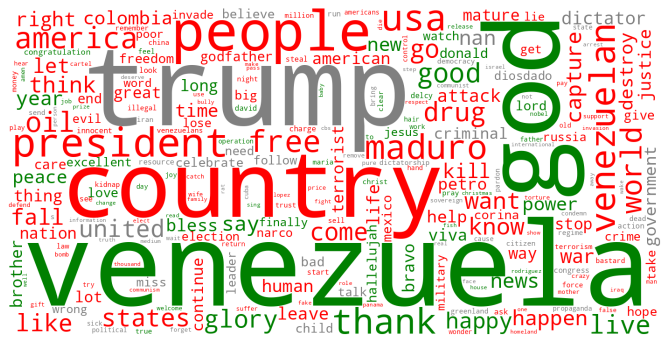


Figure 14. Female word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

4.4. Conclusion

Based on the analysis of comments from multiple YouTube Shorts and Instagram Reels related to world-at-war content, consistent gender-based engagement patterns were observed. Male users contributed a significantly higher number of comments than female users across all analysed videos, indicating higher overall participation in discussions around war-related short-form content. For both genders, the majority of comments were classified as negative or aggressive, while positive comments formed the smallest proportion of engagement, suggesting that such content generally evokes emotionally charged responses irrespective of gender. Due to their higher participation volume, male users appear more strongly influenced by exposure to this content, as reflected in the larger absolute number of negative comments; however, this reflects engagement intensity rather than inherent emotional tendency. Although a small subset of videos exhibited relatively higher positive engagement, such instances were rare, indicating that positive responses to war-related short-form content remain limited compared to neutral or negative reactions across the dataset.

5. LOCATION-BASED ANALYSIS

Palak Agarwal (230726)

The analysis looked at YouTube comments from different **locations**. It used news channels from **Venezuela** (NTN24), from the **USA** (USA Today), and from **global news** channels. Comments were taken from videos about the **USA–Venezuela** situation. Each comment was checked for **tone** and placed into one of three types: **aggressive**,

calm, or **neutral**. This helped show how people from different regions reacted to the same issue.

5.1. Venezuelan Local News:

5.1.1. Tone Distribution

The Venezuelan local news dataset contains the following tone distribution:

Table 10. Tone distribution for Venezuelan local news comments

Tone	Count	Percentage
Neutral	3089	51.09%
Calm	2790	46.13%
Aggressive	168	2.78%

Most of the comments on Venezuelan local news videos are **neutral**, followed closely by **calm** comments. Only a very small fraction of the comments (approximately **2.8%**) are classified as **aggressive**, indicating an overall low level of hostility in the discussion.

5.1.2. Word Cloud for Venezuelan Local News (Aggression-Focused Interpretation)

The word cloud for Venezuelan local news comments reveals a combination of emotionally charged and socially reflective language. High-frequency terms such as *criminals*, *tyrant*, *lie*, *suffer*, *attack*, and *die* indicate that a section of viewers expresses frustration, anger, and emotional distress related to perceived injustice, repression, and collective suffering.

From an **aggression-focused perspective**, the presence of harsh and harm-associated vocabulary reflects confrontational responses and heightened emotional reactions among some commenters. These terms suggest that aggression is often rooted in lived experiences and strong emotional responses rather than random hostility.

At the same time, the word cloud also highlights a substantial presence of calmer and socially conscious words such as *thank*, *God*, *glory*, *bless*, *happy*, *justice*, and *gift*. This indicates that many users frame their comments around humanitarian concerns, accountability, and moral reflection rather than direct verbal attacks.

Overall, the word cloud illustrates a **mixed but balanced comment environment**. While aggressive language signals emotional strain and anger, the dominant discourse remains reflective and issue-oriented, emphasizing human rights, awareness, and calls for responsible action rather than unchecked hostility.

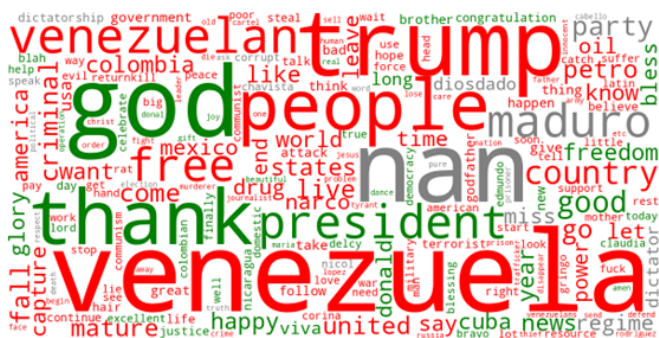


Figure 15. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

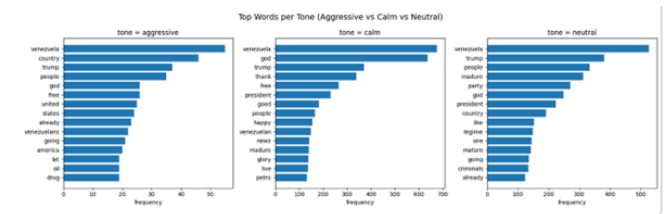


Figure 16. Top words per tone (Aggressive vs Calm vs Neutral)

5.1.3. ABC Model (Venezuelan Local News Comments)

A (Antecedent): The antecedent consists of exposure to news related to a severe political and humanitarian crisis involving leadership decisions, government actions, and widespread public suffering. Repeated narratives around injustice, punishment, and hardship act as emotional triggers, generating concern, frustration, and moral judgement among viewers.

B (Behavior): The behavioral response is reflected in the tone of user comments. Most viewers respond in a **calm or neutral** manner, engaging in discussion using language related to people, rights, justice, healthcare, and media reporting. A smaller segment of users exhibits **aggressive behavior**, employing harsh or emotionally charged vocabulary when expressing anger, blame toward leadership, or frustration over ongoing suffering.

C (Consequence): The presence of aggressive comments can increase emotional intensity and polarization within discussion threads, potentially reducing constructive dialogue. However, since calm and neutral responses dominate the conversation, the overall discourse remains largely focused on understanding the issue and expressing concern rather than escalating hostility.

5.2. US Local News:

5.2.1. Tone Distribution

The US local news dataset contains the following tone distribution:

Table 11. Tone distribution for US local news comments

Tone	Count	Percentage
Neutral	1239	62.17%
Calm	682	34.22%
Aggressive	72	3.61%

Most comments on US local news videos are **neutral**, indicating discussion-oriented reactions to the issue. Only a very small share of comments (approximately **3.6%**) are classified as **aggressive**, suggesting limited emotional escalation in the overall conversation.

5.2.2. Word Cloud for US Local News (Aggression-Focused Interpretation)

The word cloud for US local news comments is dominated by terms related to global politics, conflict, and national interests. High-frequency words such as *war*, *country*, *president*, *world*, *America*, *China*, and *Russia* indicate that users primarily frame the discussion around international relations and power dynamics rather than personal or local experiences.

From an **aggression-focused perspective**, words such as *kill*, *attack*, *criminal*, *dictator*, and *illegal* reflect anger, fear, and security-related concerns associated with war and geopolitical conflict. However, these aggressive terms are not dominant, suggesting that heightened hostility is limited to a smaller subset of commenters.

At the same time, calmer and value-oriented words such as *good*, *hope*, *care*, *help*, *bless*, and *thank* appear frequently, indicating expressions of concern, support, and moral reflection. Overall, the word cloud reflects a **mixed but balanced discussion environment**,

where emotional reactions are present but do not dominate the overall discourse.



Figure 17. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

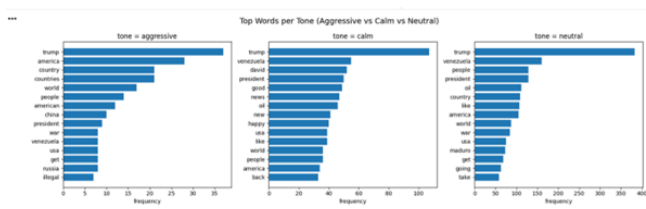


Figure 18. Top words per tone (Aggressive vs Calm vs Neutral)

5.2.3. ABC Model (US Local News Comments)

A (Antecedent): The antecedent involves exposure to US local news covering international conflict, war, and political decisions involving multiple countries and leaders. The reporting is largely factual and event-focused, encouraging viewers to think in terms of global security, power dynamics, and national interests rather than direct personal suffering.

B (Behavior): The behavioral response is primarily **neutral or calm**, with users engaging in discussion around news updates, countries, political leaders, and possible outcomes. Informational and opinion-based language dominates the comments. A smaller subset of users exhibits **aggressive behavior**, using strong vocabulary related to violence, crime, or blame, but such responses remain relatively limited.

C (Consequence): As calm and neutral comments dominate, the overall discussion remains controlled and informative. While aggressive comments occasionally introduce tension, they do not overwhelm the conversation. Consequently, the comment sections support analysis and debate rather than emotionally charged or hostile exchanges.

5.3. Global News:

5.3.1. Tone Distribution

The global news dataset contains the following tone distribution:

Table 12. Tone distribution for global news comments

Tone	Count	Percentage
Neutral	2595	60.1%
Calm	1478	34.2%
Aggressive	247	5.7%

In global news comments, most users express their views in a **neutral** tone, followed by **calm** responses, while **aggressive** comments form the smallest group. This indicates that most users discuss global issues in a measured manner without strong emotional expression, and only a small proportion of users display hostile or negative reactions.

5.3.2. Word Cloud for Global News (Aggression-Focused Interpretation)

The global news word cloud indicates that discussions are primarily centered on international politics and world affairs. High-frequency terms such as *Venezuela*, *Trump*, *America*, *Russia*, *world*, *oil*, *war*, *president*, *country*, and *USA* suggest that users focus largely on governments, political leaders, conflicts, and relations between nations.

From an **aggression-focused perspective**, words such as *kill*, *attack*, *invade*, and *drugs* appear, reflecting expressions of anger or fear related to war and political actions. However, these aggressive terms do not dominate the discussion, indicating that hostility is limited to a smaller group of users.

At the same time, calm and positive words such as *peace*, *good*, *thank*, *hope*, and *happy* are also present, suggesting that many users seek peaceful solutions or engage in opinion-sharing without hostility. Overall, the global news comment section reflects a **balanced discussion environment**, where global political discourse dominates and aggressive language remains limited.

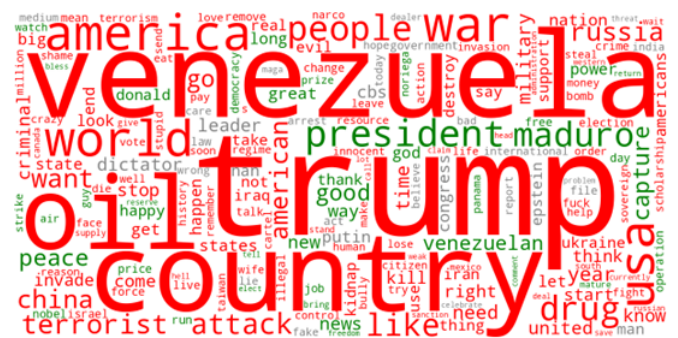


Figure 19. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

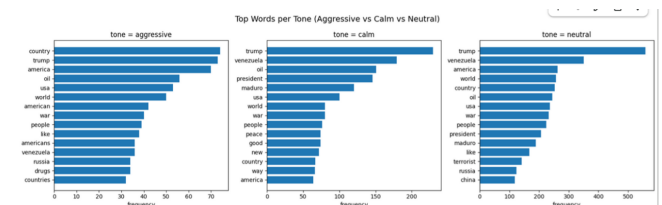


Figure 20. Overall word cloud (Aggressive = Red, Calm = Green, Neutral = Gray)

5.3.3. ABC Model (Global News Comments)

A (Antecedent): The antecedent involves exposure to news related to international conflict, sanctions, oil politics, global leadership, and power struggles between nations. These themes encourage viewers to focus on global political dynamics rather than personal or local experiences.

B (Behavior): The behavioral response is largely **neutral or calm**, with users employing language related to countries, political leaders, foreign policy, and global consequences. A smaller proportion of users exhibit **aggressive behavior**, primarily through blaming or criticizing political figures or nations.

C (Consequence): Because neutral and calm tones dominate, discussions remain mostly respectful and informative. Although aggressive comments occasionally appear, they do not disrupt the overall conversation flow. As a result, comment sections support analysis and opinion-sharing rather than emotionally charged or hostile exchanges.

5.4. Comparison of Tone Across Venezuelan, US, and Global Comments

Aggressive Tone: Venezuelan local news comments exhibit the **lowest proportion of aggressive tone (2.78%)**, followed by US local comments at **3.61%**, while global news comments show the **highest aggressive share (5.70%)**. This pattern suggests that Venezuelan users tend to express concern with minimal hostility, US discussions involve limited argumentative behavior, and global discussions reflect greater disagreement due to the presence of multiple international viewpoints.

Calm Tone: The **highest calm tone** is observed in Venezuelan comments (**46.13%**), whereas US (**34.22%**) and global (**34.20%**) comments display similar and comparatively lower calm levels. This indicates that Venezuelan viewers, due to their closer connection to the issue, respond in a more reflective and empathetic manner.

Neutral Tone: Neutral comments are most prevalent in US local news (**62.17%**), followed by global news (**60.10%**), and are lowest in Venezuelan local news (**51.09%**). This suggests that US and global audiences engage more through factual or geopolitical interpretation, while Venezuelan users show greater emotional involvement, resulting in reduced neutrality.

5.5. Comparison of ABC Model (Venezuelan vs US vs Global Comments)

Antecedent:

- Venezuelan users react to news involving direct human suffering and leadership-related issues.
- US users respond mainly to information focused on international conflict and political decisions.
- Global users view the issue as a broader geopolitical event involving multiple countries and global power dynamics.

Behavior:

- Venezuelan comments show higher levels of calm and emotional expression, with very low aggression.
- US comments are mostly neutral and discussion-based, showing limited emotional intensity.
- Global comments display mixed reactions and the highest aggressive behavior due to diverse international viewpoints.

Consequence:

- Venezuelan discussions remain empathetic and focused on human impact.
- US comment sections stay largely informative and oriented toward debate and analysis.
- Global discussions become more argumentative and politically divided compared to the other two groups.

5.6. Conclusion

Overall, comments from Venezuelan local news tend to be more **calm** and emotionally aware, reflecting a closer personal connection to the issue. US local news comments are predominantly **neutral** and analytical, focusing on factual discussion and geopolitical interpretation. In contrast, global news comments are more **politically argumentative**, which contributes to a relatively higher presence of aggressive tone.

6. TEAM

Neuronauts

Akshat Garg : 230096

Chirag Goyal : 230335

Palak Agarwal : 230726

Muskan Yadav : 230673